

3

ENERGY AND CONSERVATION LAWS

CHAPTER OUTLINE

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| 3.1 Conservation Laws | 3.6 Collisions: An Energy Perspective |
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Vern J. Ostdiek
Donald J. Bord



3.1 Conservation Laws

- The most fundamental ideas we have in physics are conservation laws.
- Statements telling us that some quantity does not change.
- One example is the law of conservation of mass.
- A **system** is defined as a collection of objects we decide to treat at one time.
- The total mass in an isolated system is constant.

3.1 Conservation Laws

A simple example of this law is given by aerial refueling. If we ignore the small amount of fuel that both aircraft consume during the refueling, this can be regarded as an isolated system.

According to law of conservation of mass the total mass of fuel dispensed by the tanker equals the total mass of fuel gained by the other aircraft.



3.1 Conservation Laws

This fact may be useful because if the fuel gauge on one of the receiving aircraft is faulty, the crew can determine how much fuel that aircraft was given using the conservation of mass.

The total mass of fuel unloaded from the tanker *minus* The total mass of fuel given to the other aircraft in the formation.



3.2 Linear Momentum

- Linear momentum is often referred to simply as *momentum*.
- It is a vector quantity (because velocity is a vector).
- Its SI unit of measure is the kilogram-meter/second (kg-m/s).

DEFINITION

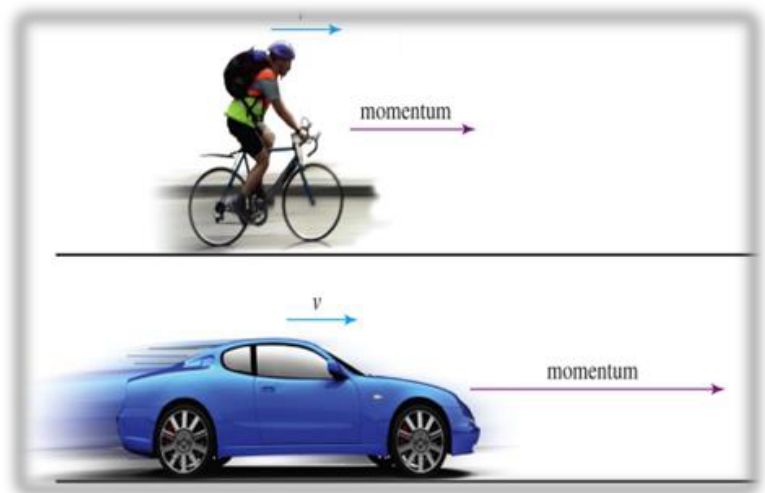
Linear Momentum: The mass of an object times its velocity. Linear momentum is a vector.

$$\text{linear momentum} = mv$$

3.2 Linear Momentum

Linear momentum incorporates both mass and motion.

- Anything that is stationary has zero momentum.
- The faster a body moves, the larger its momentum.
- A heavy object moving with a certain velocity has more momentum than a light object moving with the same velocity.

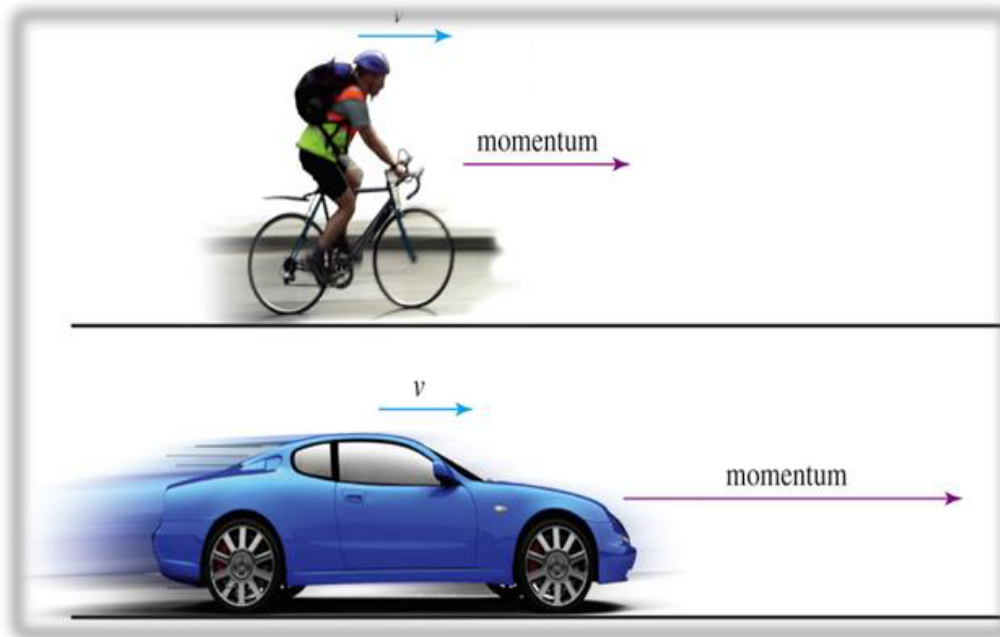


3.2 Linear Momentum

Linear momentum incorporates both mass and motion.

cyclist: $mv = 80 \text{ kg} \times 10 \text{ m/s} = 800 \text{ kg}\cdot\text{m/s}$

car: $mv = 1,200 \text{ kg} \times 10 \text{ m/s} = 12,000 \text{ kg}\cdot\text{m/s}$



3.2 Linear Momentum: Newton's Second Law, Revisited

Newton's second law of motion was originally stated using linear momentum.

To change an object's linear momentum, a net force must act on it. The larger the force, the faster the momentum will change.

LAWS

Newton's Second Law of Motion (alternate form**):**

The net external force acting on an object equals the rate of change of its linear momentum.

3.2 Linear Momentum: Newton's Second Law, Revisited

If the mass of the object stays constant, which is true in many cases. Then:

$$\text{force} = \frac{\text{change in momentum}}{\text{change in time}}$$

$$F = \frac{\Delta(mv)}{\Delta t}$$

$$\frac{\Delta(mv)}{\Delta t} = m \frac{\Delta v}{\Delta t} = ma \quad (\text{if mass is constant})$$

3.2 Linear Momentum: Newton's Second Law, Revisited

Either form of the second law could be used for cars, airplanes, baseballs, and so on.

For rockets and similar things with changing mass, only the alternate form should be used.

3.2 Linear Momentum: Newton's Second Law, Revisited

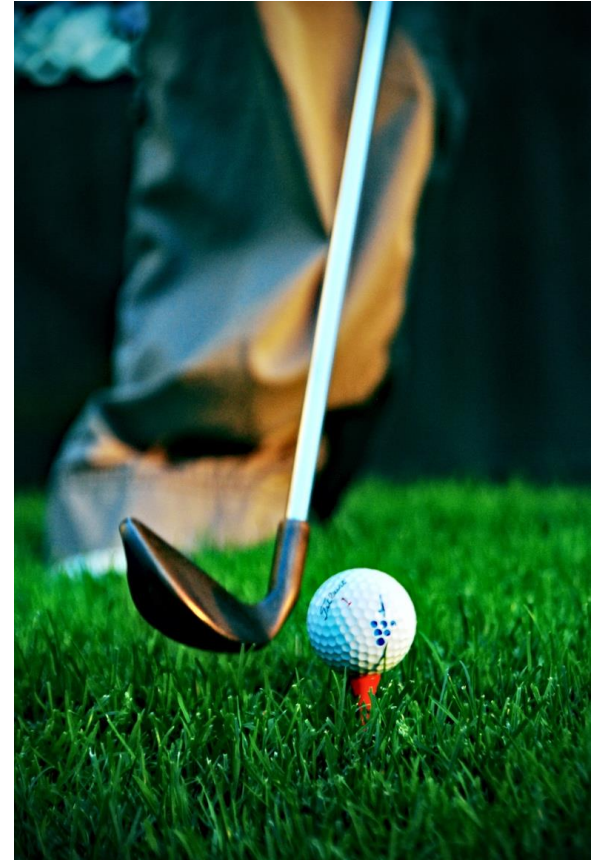
$$\Delta(mv) = F\Delta t$$

The quantity on the right side is called the *impulse*. The impulse equals the change in momentum.

The same change in momentum can result from a small force acting for a long time *or* a large force acting for a short time.

Example 3.1

Let's estimate the average force on a golf ball as it is driven off a tee. The ball's mass is 0.045 kg , and it leaves the club head with a horizontal speed of, say, 50 m/s (110 mph). High-speed photographs indicate that the contact time is about 5 milliseconds (0.005 s). (We will ignore any spin imparted to the ball upon impact.)



Example 3.1, continued

$$\begin{aligned}\Delta(mv) &= \text{momentum afterwards} = 0.045 \text{ kg} \times 50 \text{ m/s} \\ &= 2.25 \text{ kg-m/s}\end{aligned}$$

$$\begin{aligned}F &= \frac{\Delta(mv)}{\Delta t} = \frac{2.25 \text{ kg-m/s}}{0.005 \text{ s}} \\ &= 450 \text{ N} = 101 \text{ lbs}\end{aligned}$$

3.2 Linear Momentum: Conservation of Linear Momentum (Collisions)

LAWS

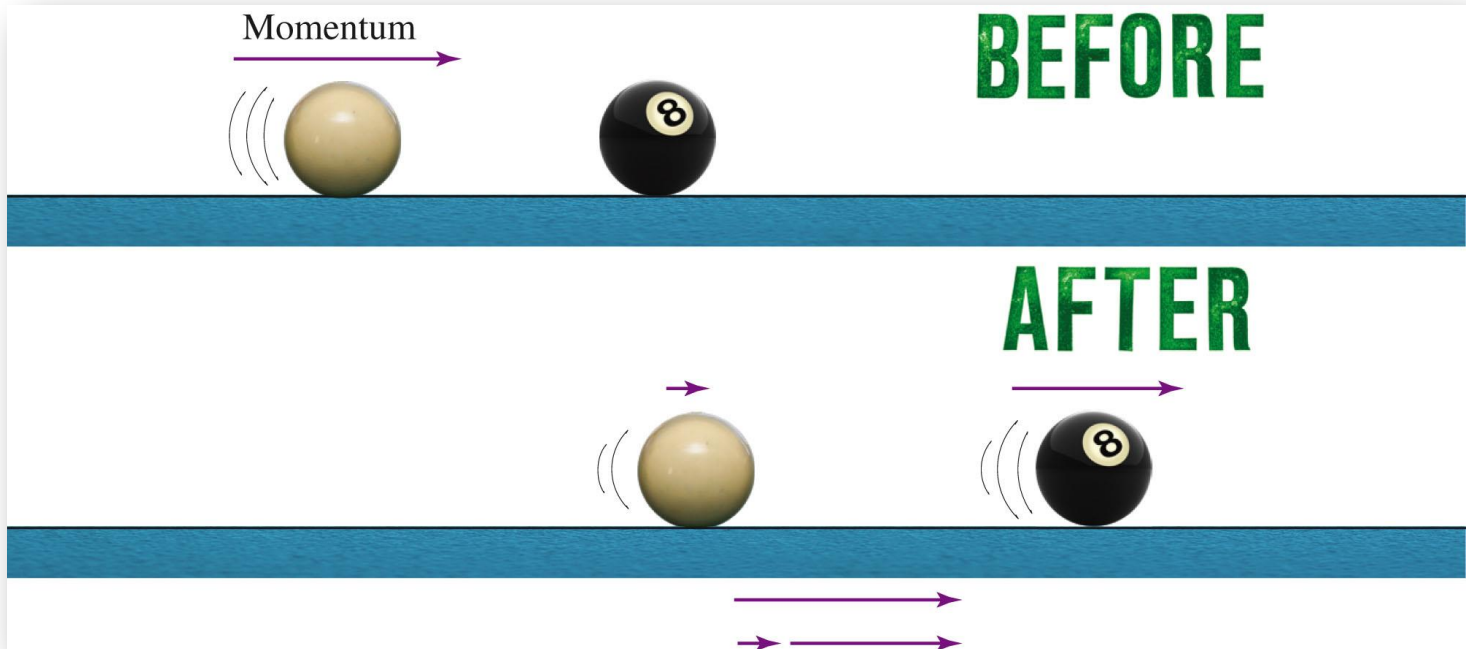
Law of Conservation of Linear Momentum: The total linear momentum of an isolated system is constant.

For this law, an isolated system means that there are *no outside forces* causing changes in the linear momenta of the objects inside the system.

$$F = 0 = \frac{\Delta p}{\Delta t} \Rightarrow \Delta p = 0$$

3.2 Linear Momentum: Conservation of Linear Momentum (Collisions)

If the cue ball collides with another ball initially at rest, its momentum is changed (decreased).



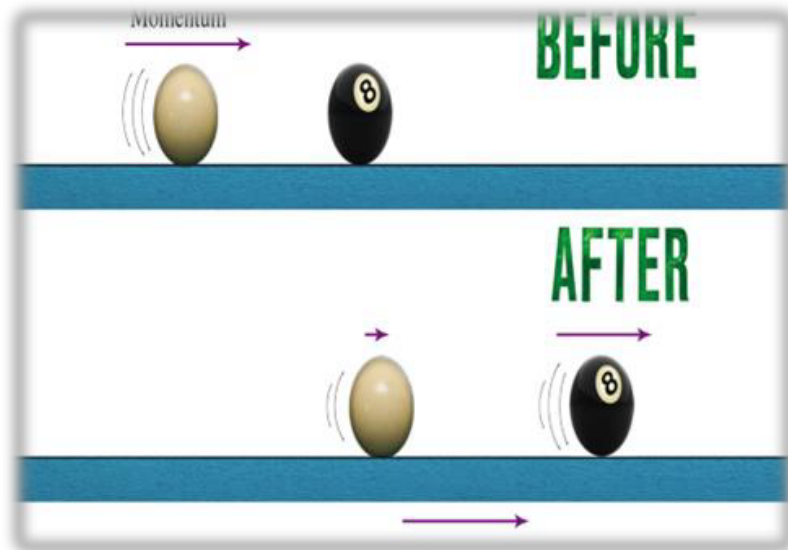
This change occurs because of an interaction with another object *in the system*.

3.2 Linear Momentum: Conservation of Linear Momentum (Collisions)

If the cue ball collides with another ball initially at rest, its momentum is changed (decreased).

The momentum of the ball with which it collides is increased, but the total linear momentum of the system remains the same

If someone put a hand on the table and stopped the cue ball, the system would no longer be isolated, and we couldn't assume that its total linear momentum was constant.



3.2 Linear Momentum: Conservation of Linear Momentum (Collisions)

The most important use of the law of conservation of linear momentum is in the analysis of collisions.

During any collision, the colliding objects exert equal and opposite forces on each other that cause them to accelerate (in opposite directions).

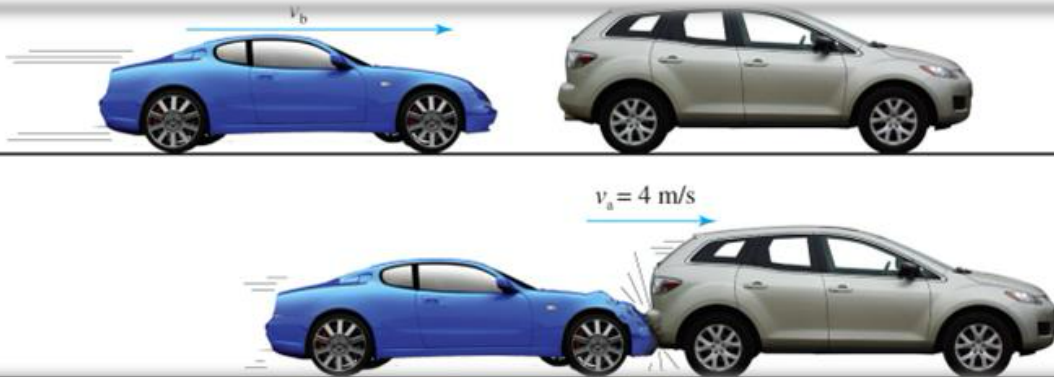
LAWS

The total linear momentum of the objects in a system before the collision is the same as the total linear momentum after the collision.

$$\text{total } mv \text{ before} = \text{total } mv \text{ after}$$

Example 3.2

A 1,000-kilogram automobile (car 1) runs into the rear of a stopped car (car 2) that has a mass of 1,500 kilograms. Immediately after the collision, the cars are hooked together, and their subsequent speed is estimated to be 4 m/s. What was the speed of car 1 just before the collision?



Example 3.2, continued

$$\text{total}(mv)_{\text{before}} = \text{total}(mv)_{\text{after}}$$

$$(mv)_{\text{before}} = m_1 \times v_{\text{before}} = (1,000 \text{ kg}) \times v_{\text{before}}$$

$$\begin{aligned}(mv)_{\text{after}} &= (m_1 + m_2) \times v_{\text{after}} \\ &= (1,000 \text{ kg} + 1,500 \text{ kg}) \times 4 \text{ m/s} \\ &= 2,500 \text{ kg} \times 4 \text{ m/s} = 10,000 \text{ kg-m/s}\end{aligned}$$

$$1,000 \text{ kg} \times v_{\text{before}} = 10,000 \text{ kg-m/s}$$

$$v_{\text{before}} = \frac{10,000 \text{ kg-m/s}}{1,000 \text{ kg}} = 10 \text{ m/s}$$

3.2 Linear Momentum: Conservation of Linear Momentum (Collisions)

How do rockets work?

Rockets and jets give momenta to the ejected exhaust gases. They in turn gain momentum in the forward direction.

For a rocket with no external forces on it, the increase in momentum during each second will depend on how much gas is ejected—which equals the mass of fuel burned—and on the speed of the gas.



3.3 Work: The Key to Energy

One could say that every interaction in our universe involves a transfer of energy or a transformation of energy from one form to another.

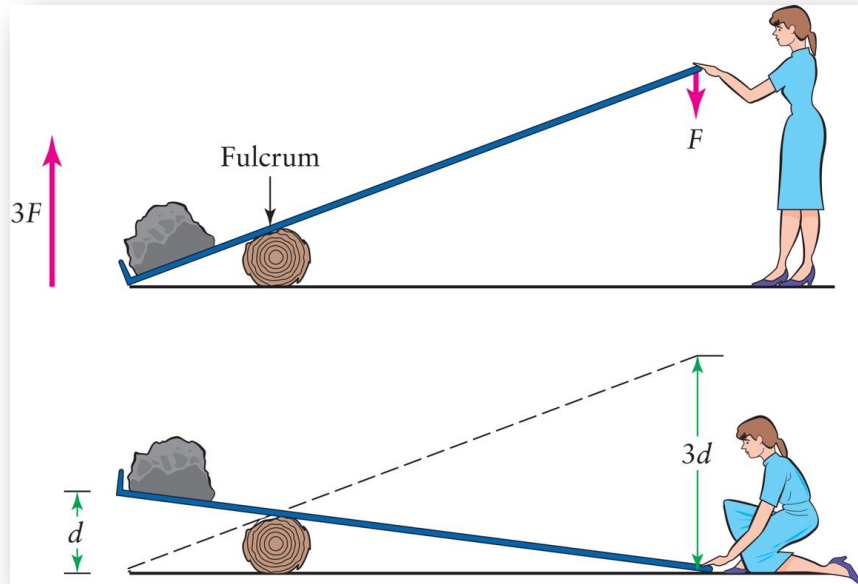
The concept of energy is a bit difficult to understand because there is no simple way to define it.

As an aid, we will first introduce work, a physical quantity that is quite basic and that gives us a good foundation for understanding energy.

3.3 Work: The Key to Energy

Let's say that you use a lever to raise a heavy rock

place the fulcrum close to the rock, you find that a small, downward force on your end results in a larger, upward force on the rock.



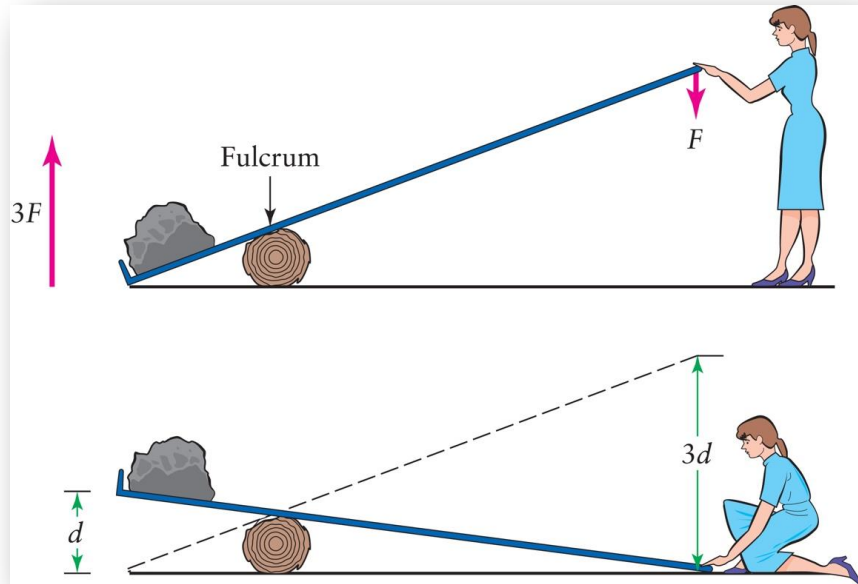
3.3 Work: The Key to Energy

However, the distance your end of the lever moves as you push it down is correspondingly larger than the distance the rock is raised.

There must be some connection between force and distance.

input = output

$$(F)(3d) = (3F)(d)$$



3.3 Work: The Key to Energy

By measuring the forces and distances, we find that dividing the larger force by the smaller force gives the same number (or ratio) as dividing the larger distance by the smaller distance.

In particular,

$$\frac{F \text{ on the left end}}{F \text{ on the right end}} = \frac{d \text{ right end moves}}{d \text{ left end moves}}$$

Rearranging the equation to get:

$$(F \text{ on left}) \times (d \text{ left moves}) = \\ (F \text{ on right}) \times (d \text{ right moves})$$

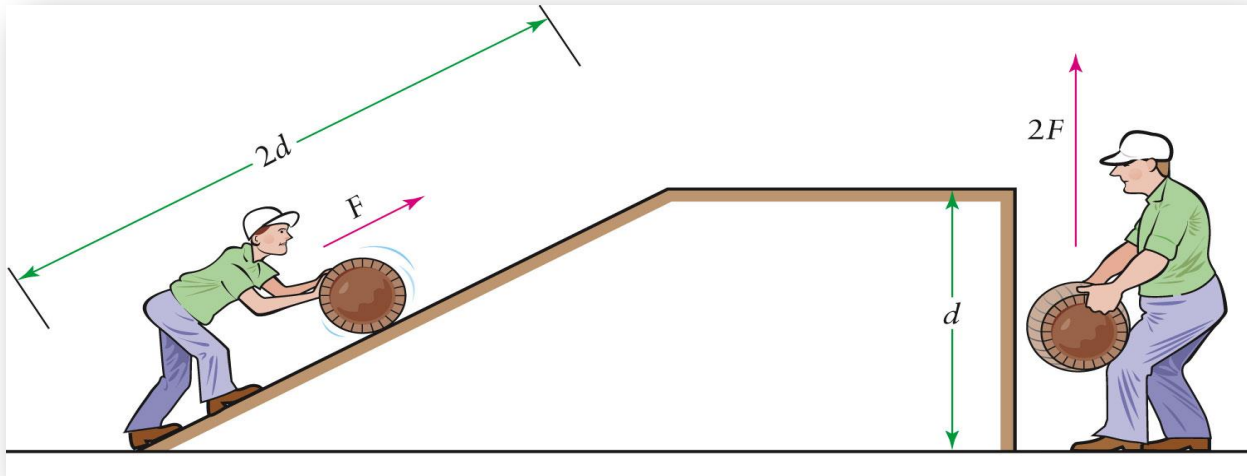
$$F_{\text{left}} d_{\text{left}} = F_{\text{right}} d_{\text{right}}$$

3.3 Work: The Key to Energy

In other words, even though the two forces and the two distances are different, the quantity *force times distance* has the same value for both ends of the lever.

3.3 Work: The Key to Energy

Similarly,



$$(F \text{ lifting}) \times \text{height} = (F \text{ rolling}) \times (\text{ramp length})$$

$$F_{\text{lifting}} d_{\text{lifting}} = F_{\text{rolling}} d_{\text{rolling}}$$

3.3 Work: The Key to Energy

The quantity *force times distance* is obviously a useful way of measuring the “size” of a task. It is called **work**.

DEFINITION

Work: The force that acts times the distance moved in the direction of the force.

$$\text{work} = Fd$$

- Work is just that part of the force in the direction of the displacement.
- Work is not a vector — it's a scalar.

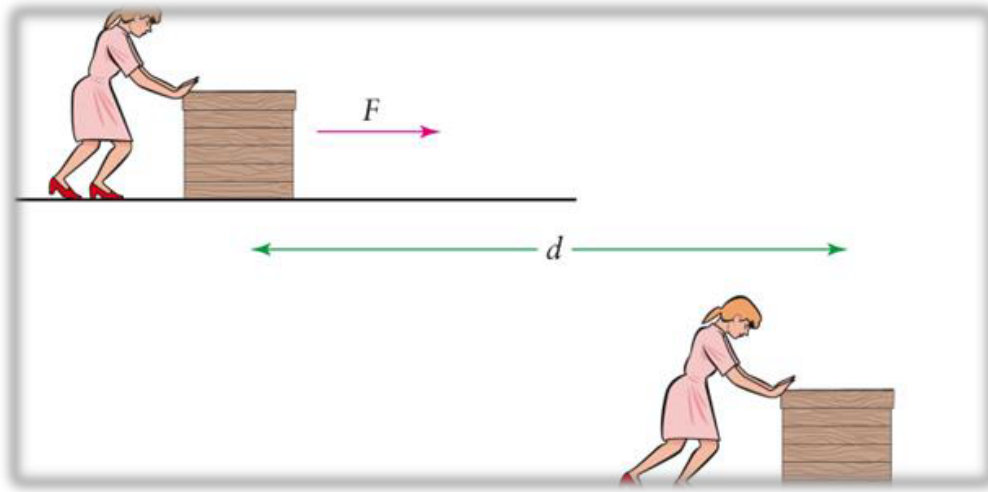
3.3 Work: The Key to Energy

The following table lists the common Work units of measure, in both the *metric* and the *English systems* and their abbreviations.

Physical Quantity	Metric Units	English Units
Work	joule (J) [SI unit]	foot-pound (ft-lb)
	erg	British thermal unit (Btu)
	calorie (cal)	
	kilowatt-hour (kWh)	

Example 3.3

Because of friction, a constant force of 100 newtons is needed to slide a box at a steady speed across a room. If the box moves 3 meters, how much work is done?



$$\text{work} = Fd = 100 \text{ N} \times 3 \text{ m} = 300 \text{ N}\cdot\text{m}$$

$$1 \text{ joule} = 1 \text{ newton-meter} = 1 \text{ newton} \times 1 \text{ meter} \rightarrow 1 \text{ J} = 1 \text{ N}\cdot\text{m}$$

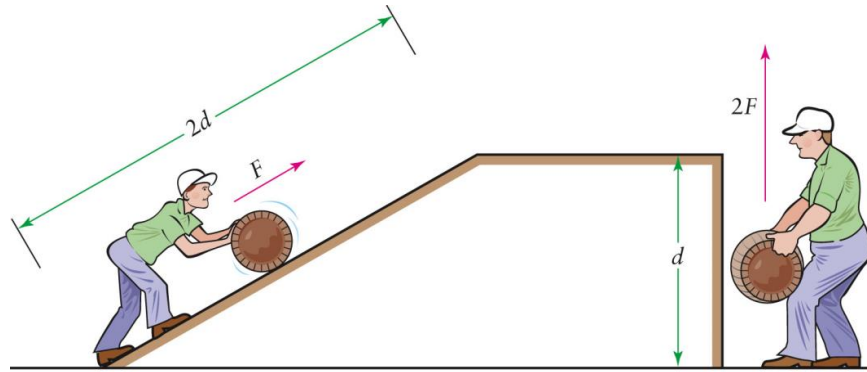
Example 3.4

Let's say that the barrel in the figure has a mass of 30 kilograms and that the height of the dock is 1.2 meters. How much work would you do when lifting the barrel?

$$\text{work} = Fd$$

$$F = W = mg = 30 \text{ kg} \times 9.8 \text{ m/s}^2 = 294 \text{ N}$$

$$\text{work} = Fd = Wd = 294 \text{ N} \times 1.2 \text{ m} = 353 \text{ J}$$



3.3 Different Forces, Different Work

There is no direction associated with work, although work may be positive or negative.

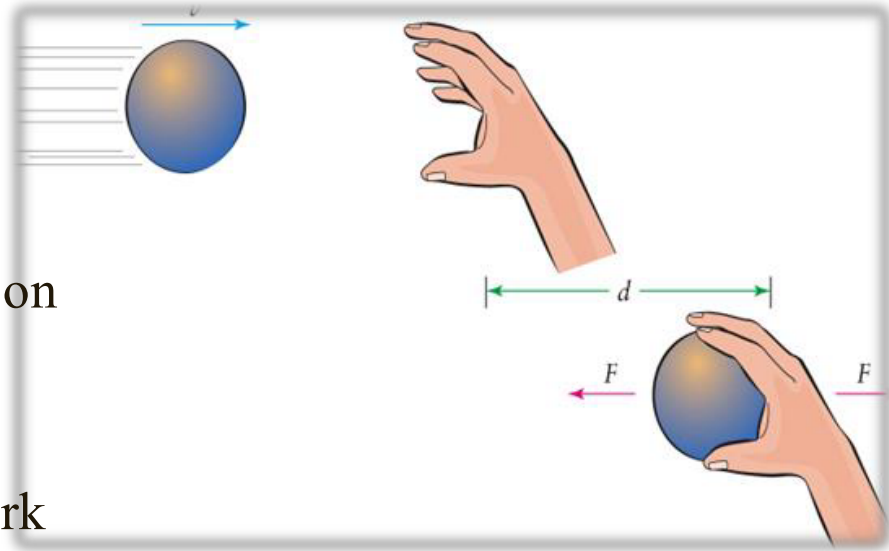
- If the force and distance are in the same direction, the force does *positive work*.
- If the force and distance are in the opposite direction, the force does *negative work*.

3.3 Different Forces, Different Work

When you throw or catch a ball, your hand exerts a force on the ball. As the ball slows down, it pushes your hand back with an equal and opposite force.

In this case:

- ✓ The ball does positive work on your hand.
- ✓ Your hand does negative work on the ball.



3.3 Different Forces, Different Work

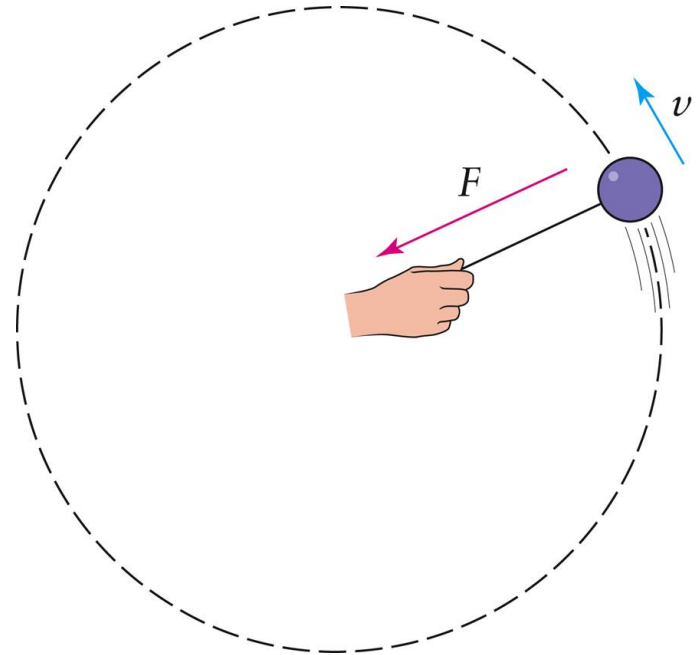
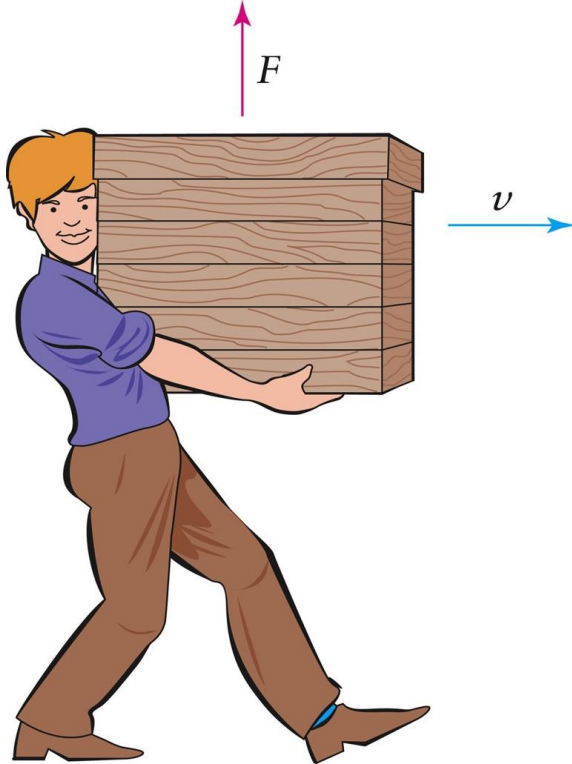
The amount of work that the ball does on your hand is equal to the amount of work that was originally done on the ball to accelerate it (ignoring the effect of air resistance).

If you allow your hand to move back as you catch the ball, the force of the ball on you will be less than if you try to keep your hand stationary.

The work that the ball will do on your hand is the same either way. Since $\text{work} = \text{force} \times \text{distance}$, the force on your hand will be smaller if the distance that the ball moves while you catch it is larger.

3.3 Different Forces, Different Work

If the force is perpendicular to the distance, the force does *no work*



Example 3.5

The force needed to accelerate a 1,000-kilogram car from 0 to 27 m/s in 10 seconds is 2,700 N. How much work is done?

$$d = \frac{1}{2}at^2$$

$$= \frac{1}{2} \times 2.7 \text{ m/s}^2 \times (10 \text{ s})^2$$

$$= 1.35 \text{ m/s}^2 \times 100 \text{ s}^2 = 135 \text{ m}$$

$$\text{work} = Fd$$

$$= 2,700 \text{ N} \times 135 \text{ m} = 364,500 \text{ J}$$

3.4 Energy

- When work is done on something, it gains **energy**.
- This energy can then be used to do work.
- The units of energy are the same as the units of work.
- Like work, energy is a scalar; it has no direction associated with it, although it can be positive or negative.

DEFINITION

Energy:

The measure of a system's capacity to do work.
which is transferred when work is done.

Abbreviated *E*.

3.4 Energy

Energy is “stored work.”

To be able to do work, an object must have energy.

3.4 Energy: Mechanical Energy- Kinetic and Potential

In mechanics, there are two main forms of energy, which we can classify under *mechanical energy*.

DEFINITION

Kinetic Energy:

Energy resulting from motion. Energy that an object has because it is moving. Abbreviated *KE*.

Anything that is moving has kinetic energy.

3.4 Energy: Mechanical Energy- Kinetic and Potential

The amount of kinetic energy an object has depends on its mass and speed. In particular,

$$KE = \frac{1}{2}mv^2$$

- The kinetic energy that an object has is equal to the work done when accelerating the object from rest.
- To determine the amount of work done when accelerating an object is to compute its kinetic energy.

Example 3.6

In Example 3.5, we computed the work that is done on a 1,000-kilogram car as it accelerates from 0 to 27 m/s. Let's compute the total kinetic energy gained by the car during its acceleration.

$$\begin{aligned} KE &= \frac{1}{2}mv^2 = \frac{1}{2} \times 1,000 \text{ kg} \times (27 \text{ m/s})^2 \\ &= 500 \text{ kg} \times 729 \text{ m}^2/\text{s}^2 = 364,500 \text{ J} \end{aligned}$$

3.4 Energy: Mechanical Energy- Kinetic and Potential

The amount of **potential energy** that a system acquires is equal to the work done to put it into a given orientation, position, or configuration relative to other objects.

DEFINITION

Potential Energy:

Energy resulting from an object's position or orientation.

When an object is lifted, it is given potential energy.

3.4 Energy: Mechanical Energy- Kinetic and Potential

The work that is done against the force of gravity, is called **gravitational potential energy**.

If you lift an object, you must apply a force at least equal to the object's weight:

$$\begin{aligned} PE &= \text{work done} = \text{weight} \times \text{vertical displacement} \\ &= Wd = mgd \end{aligned}$$

$$PE = Wd = mgd \quad (\text{gravitational potential energy})$$

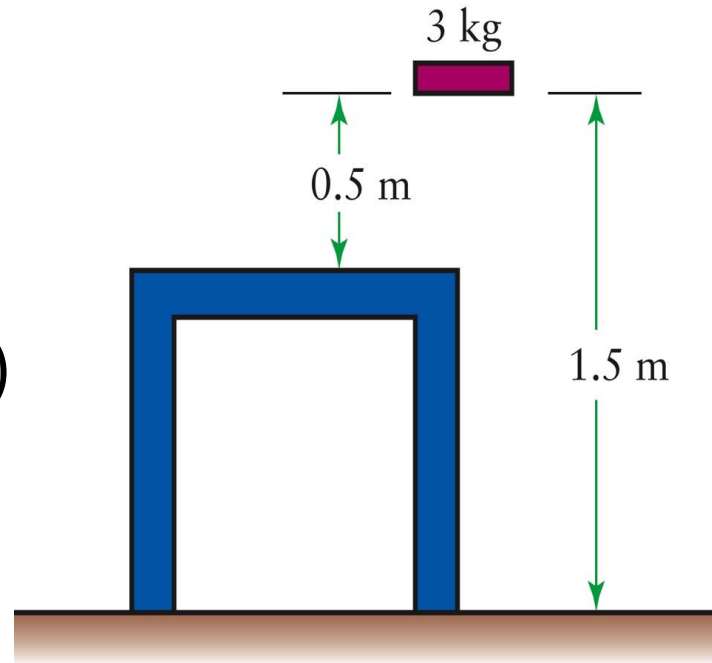
Example 3.7

A 3-kilogram brick is lifted to a height of 0.5 meters above a table. What is its potential energy relative to the table top?

$$PE = mgd$$

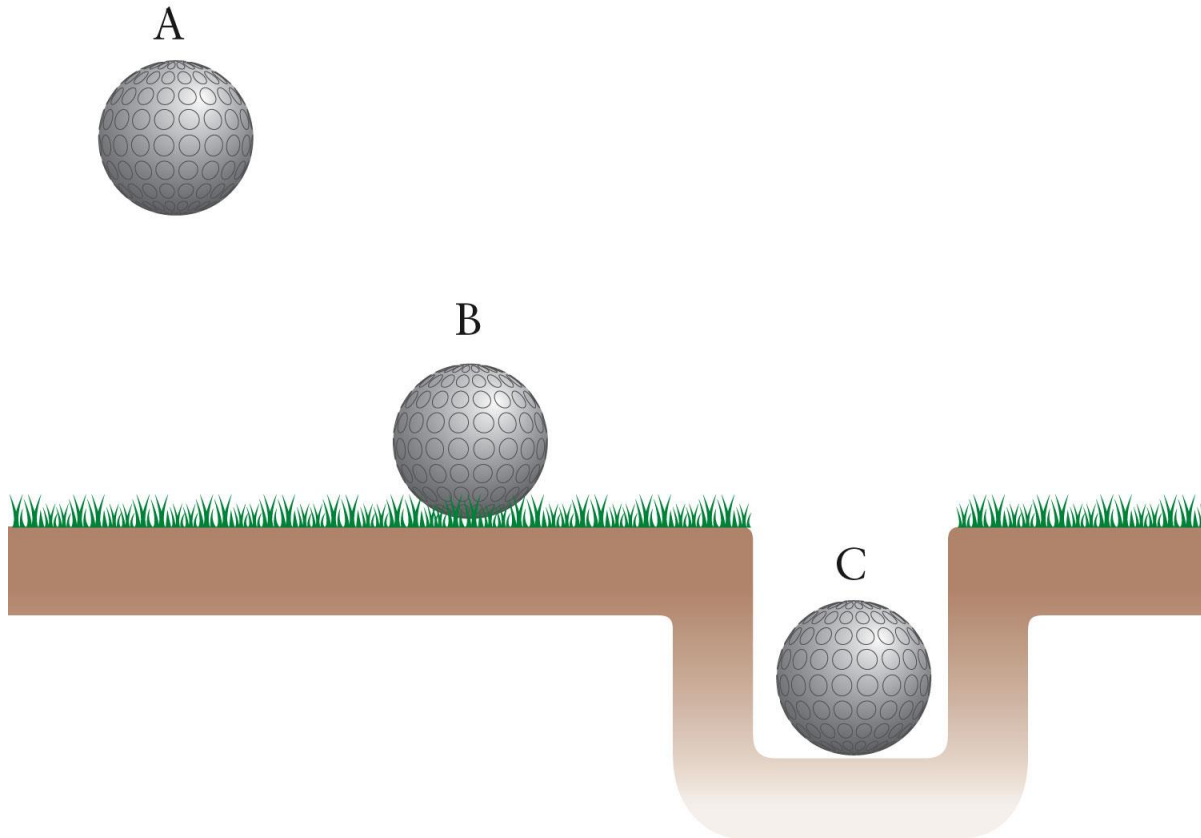
$$PE = 3 \text{ kg} \times 9.8 \text{ m/s}^2 \times 0.5 \text{ m}$$
$$= 14.7 \text{ J} \quad (\text{relative to the table})$$

$$PE = 3 \text{ kg} \times 9.8 \text{ m/s}^2 \times 1.5 \text{ m}$$
$$= 44.1 \text{ J} \quad (\text{relative to the floor})$$



3.4 Energy: Mechanical Energy- Kinetic and Potential

Where you say an object has zero PE is arbitrary. We only care about the change.



3.4 Energy: Mechanical Energy- Internal energy

- **Internal energy** of a substance is the total energy of all the atoms and molecules in the substance.
- To raise something's temperature, to melt a solid, or to boil a liquid all require increasing the internal energy of the substance.
- Internal energy decreases when a substance's temperature decreases, when a liquid freezes, or when a gas condenses.

A hot rock has more internal energy than a cold rock.

3.4 Energy: Mechanical Energy- Internal energy

Work always results in:

A transfer of energy from one thing to another

A transformation of energy from one form to another
or both.

3.5 Conservation of Energy

Many devices in common use are simply energy converters. Some examples are listed below

Device	Energy Conversion
Lightbulb	Electrical energy to radiant energy
Car engine	Chemical energy (in fuel) to kinetic energy
Battery	Chemical energy to electrical energy
Elevator	Electrical energy to gravitational potential energy
Generator	Kinetic energy to electrical energy
Electric motor	Electrical energy to kinetic energy
Solar cell	Radiant energy (light) to electrical energy
Flute	Kinetic energy (of air) to acoustic energy (of sound wave)
Nuclear power plant	Nuclear energy to electrical energy
Hydroelectric dam	Gravitational potential energy (of water) to electrical energy

3.5 Conservation of Energy

LAWS

Law of Conservation of Energy: Energy cannot be created or destroyed, only converted from one form to another. The total energy in an isolated system is constant.

For a system to be isolated, energy cannot enter or leave it. For a mechanical system, work cannot be done on the system by an outside force, nor can the system do work on anything outside of it.

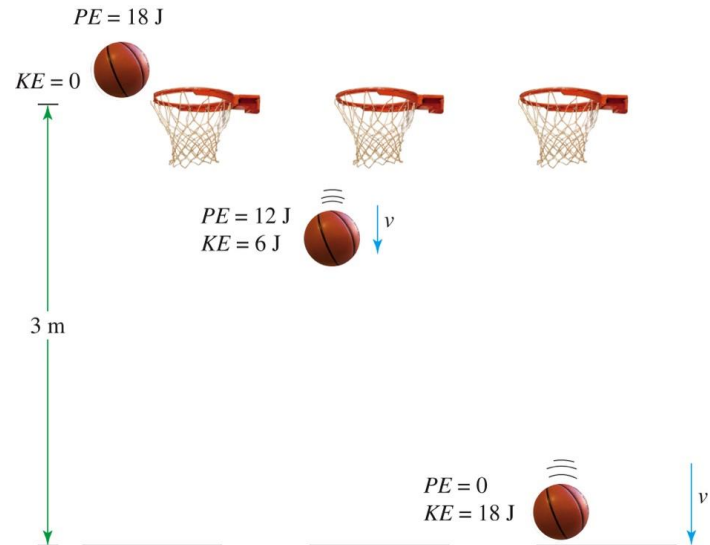
The energy of the Universe is constant

3.5 Conservation of Energy: Applications of Energy Conservation

If there is a conversion of energy in the system from one form to another, then the total energy before the conversion equals the total energy after the conversion.

total energy before = total energy after

A good example is the motion of a freely falling body. If an object is raised to a height d , it has gravitational potential energy. If it is released, the object will fall and convert its potential energy into kinetic energy. It is a continuous process

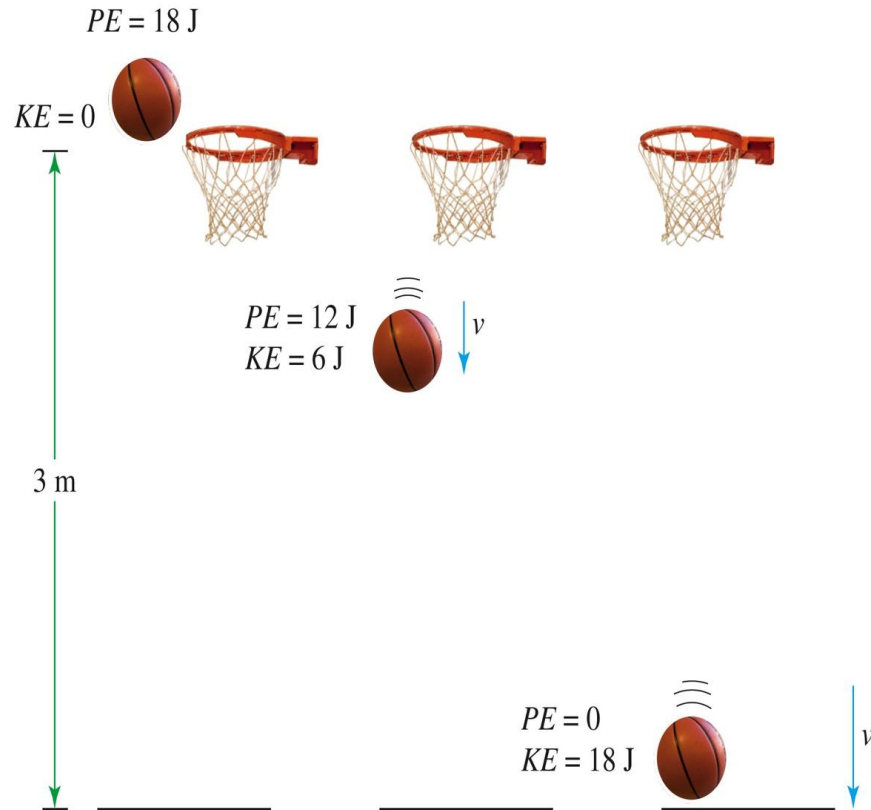


3.5 Conservation of Energy: Applications of Energy Conservation

If it is falling freely, there is no air resistance, and the only two forms of energy are kinetic and gravitational potential.

Energy conservation then means that the sum of the kinetic energy and the potential energy of the object is always the same

$$E = KE + PE = \text{constant}$$



3.5 Conservation of Energy: Applications of Energy Conservation

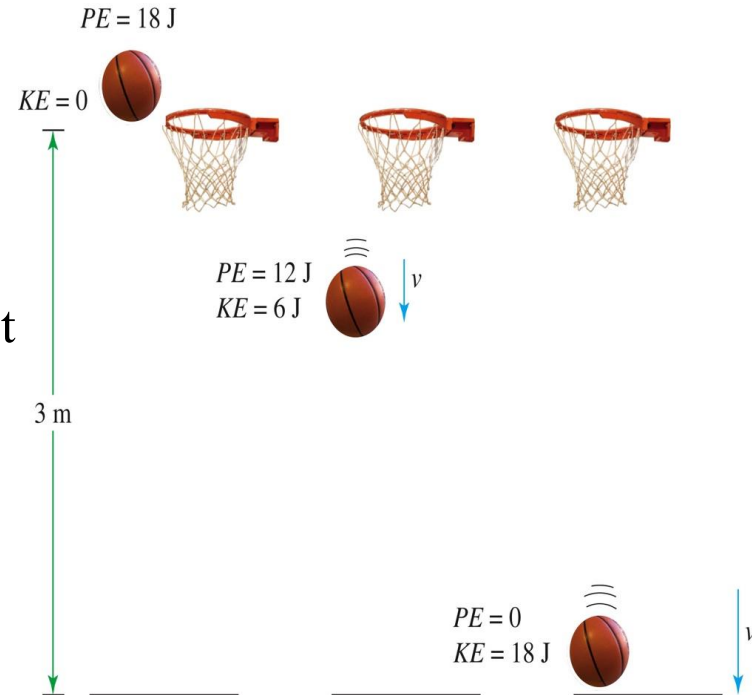
$$\begin{aligned} E_i &= KE + PE = 0 + PE \\ &= PE = mgd \quad (\text{just released}) \end{aligned}$$

$$\begin{aligned} E_f &= KE + PE = KE + 0 \\ &= KE = \frac{1}{2}mv^2 \quad (\text{just before impact}) \end{aligned}$$

$$\frac{1}{2}mv^2 = mgd$$

$$v^2 = 2gd$$

$$v = \sqrt{2gd} \quad (\text{speed after falling distance } d)$$



Example 3.9

In 2003, a man went over Horseshoe Falls, part of Niagara Falls, and survived. He is thought to be the first person to do so without the aid of any safety devices. The height of the falls is about 50 meters. Estimate the speed of the man when he hit the water at the bottom of the falls.

$$\begin{aligned} v &= \sqrt{2gd} = \sqrt{2 \times 9.8 \text{ m/s}^2 \times 50 \text{ m}} \\ &= \sqrt{980 \text{ m}^2/\text{s}^2} = 31.3 \text{ m/s} \quad (\text{about } 70 \text{ mph}) \end{aligned}$$

3.5 Conservation of Energy: Applications of Energy Conservation

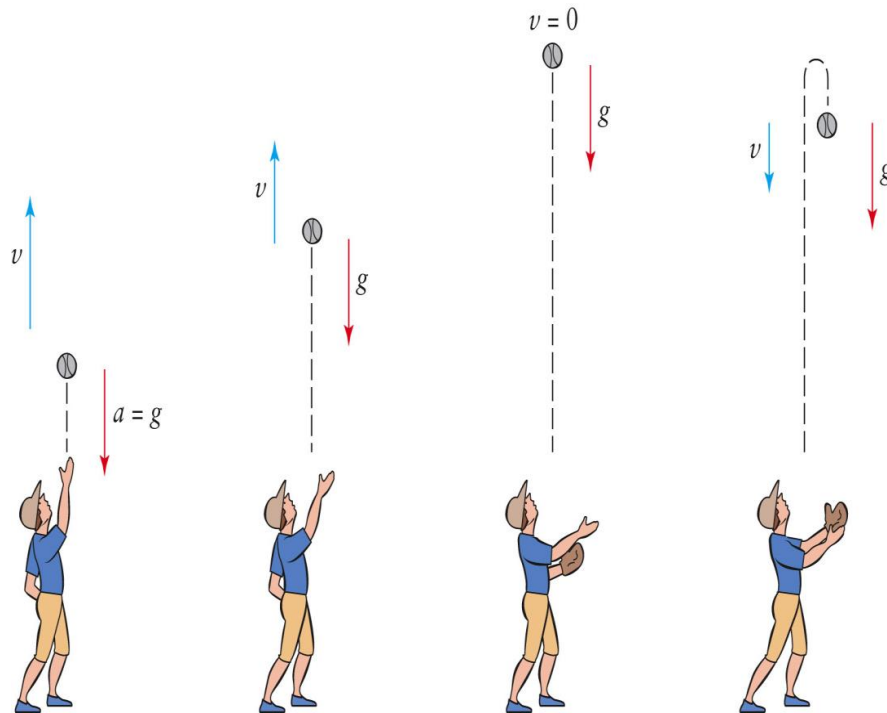
The speed of a freely falling body does not depend on its mass. It only depends on how far it has fallen (d) and the acceleration resulting from gravity.

3.5 Conservation of Energy: Applications of Energy Conservation

when an object is thrown or projected straight up:

It starts with kinetic energy, rises until all of that has been converted into potential energy, and then falls. The conservation of energy tells us that the kinetic energy that the object begins with equals its potential energy at the highest point.

$$v^2 = 2gd \rightarrow d = \frac{v^2}{2g}$$



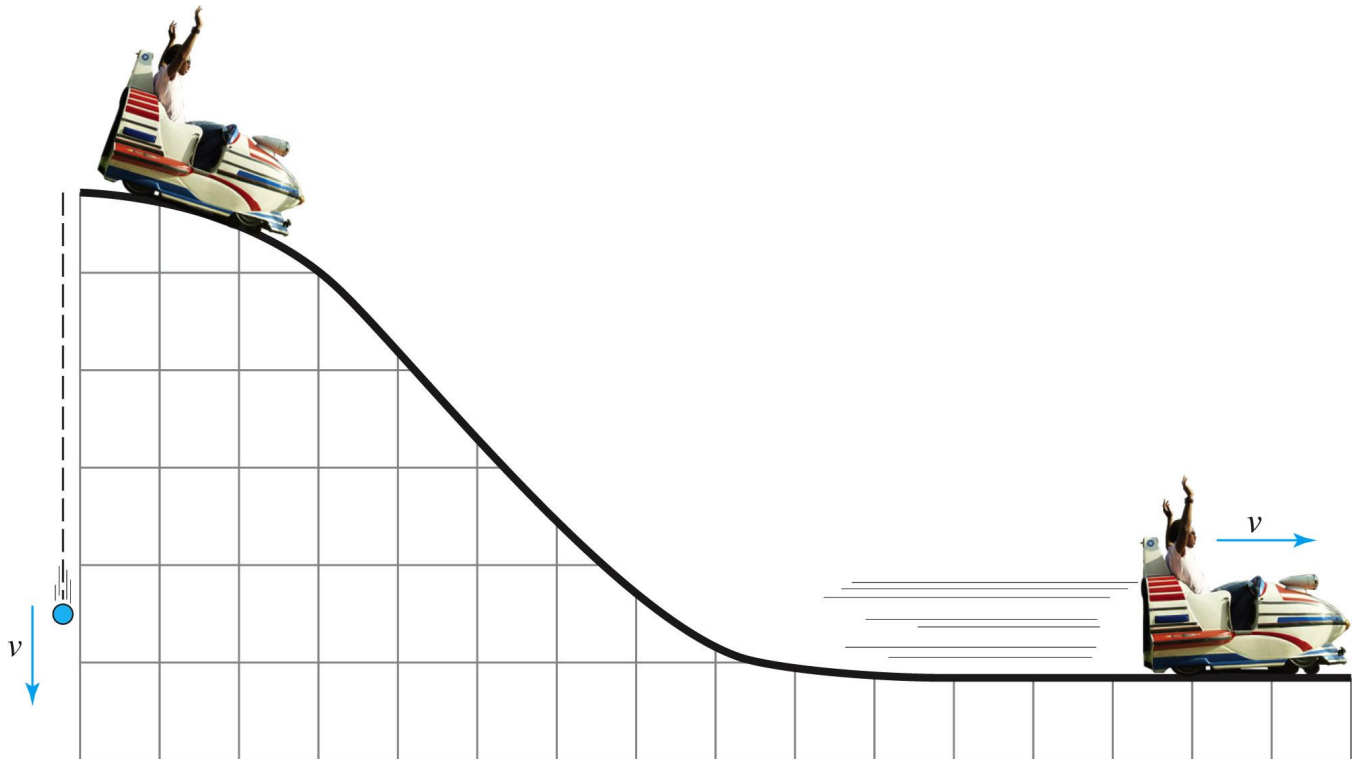
3.5 Conservation of Energy: Applications of Energy Conservation

I toss a 0.06-kg tennis ball straight up. When it leaves my hand, it has a speed of 20 m/s. Find how high the ball rises.

$$\begin{aligned} h &= \frac{v^2}{2g} = \frac{(20 \text{ m/s})^2}{2(9.8 \text{ m/s}^2)} \\ &= 20.4 \text{ m.} \end{aligned}$$

3.5 Conservation of Energy: Applications of Energy Conservation

A roller coaster starts from rest at a height d above the ground. It rolls without friction or air resistance down a hill. What is its speed when it reaches the bottom?



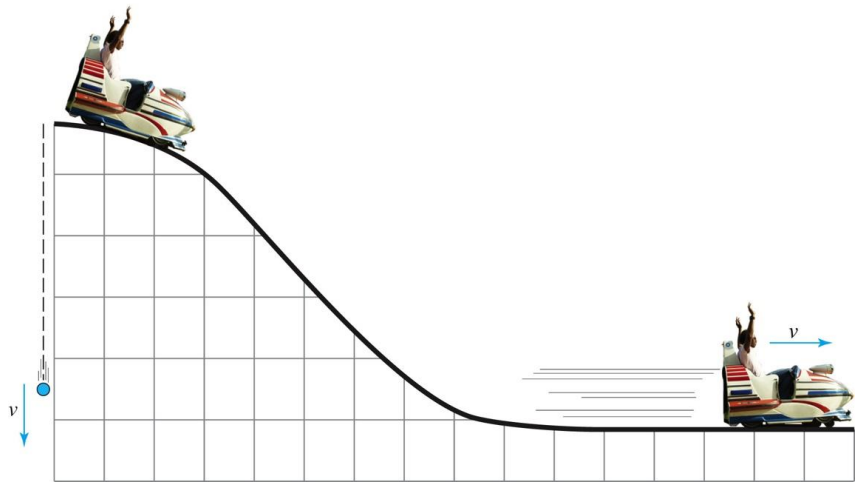
3.5 Conservation of Energy: Applications of Energy Conservation

The only forms of energy are gravitational potential energy and kinetic energy, because we assume there is no friction. The total energy, which is kinetic energy plus potential energy, is constant. The kinetic energy of the roller coaster at the bottom of the hill must equal its potential energy at the top:

$$KE \text{ (at bottom)} = PE \text{ (at top)}$$

$$\frac{1}{2}mv^2 = mgd$$

$$\rightarrow v = \sqrt{2gd}$$



3.5 Conservation of Energy: Applications of Energy Conservation

solving this problem using Newton's second law, $F = ma$, we need to know the net force that acts at every instant. The net force driving the roller coaster along its path varies as the slope of the hill changes, so it is a very complicated problem.

The principle of energy conservation allows us to easily solve a problem that we could not have solved before

3.6 Collisions: An Energy Perspective-Types of Collisions

A collision is when two objects interact to exchange energy and/or exchange momentum.

Collisions may be classified as follows:

An **Elastic Collision** is one in which the total kinetic energy of the colliding bodies after the collision *equals* the total kinetic energy before the collision.

An **Inelastic Collision** is one in which the total kinetic energy of the colliding bodies after the collision *is not equal to* the total kinetic energy before the collision. The total kinetic energy after can be greater than, or less than, the total kinetic energy before.

3.6 Collisions: An Energy Perspective-Types of Collisions

The total energy is *always* conserved in both types of collisions.

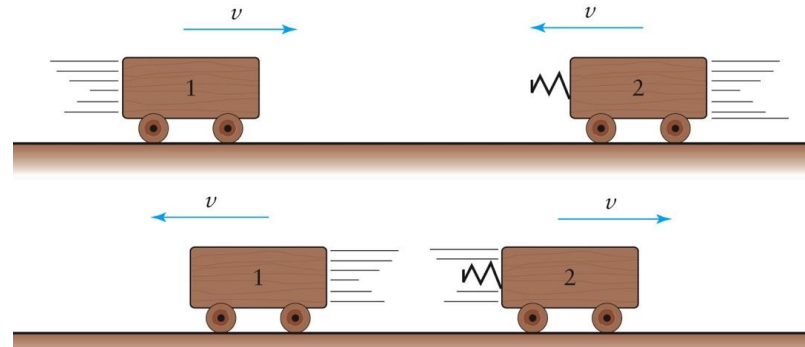
In an elastic collision, kinetic energy is conserved.

3.6 Collisions: An Energy Perspective-Types of Collisions

Two carts with the same mass m and speed v collide head on and bounce apart.

The total kinetic energy of the carts is the same before and after the collision.

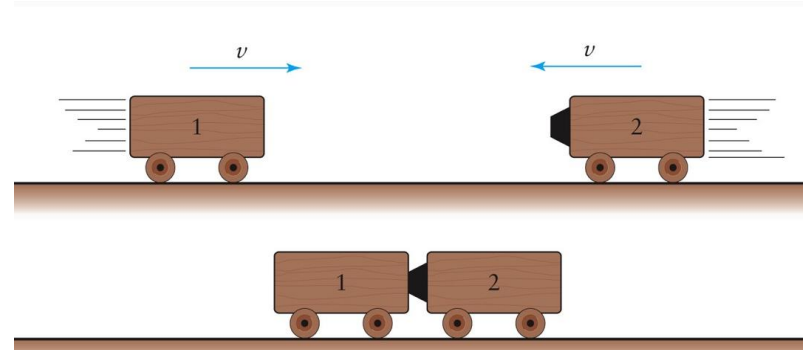
This is an elastic collision.



Two carts with the same mass m and speed v collide head on and stick together after the collision.

The kinetic energy of the carts after the collision is zero.

This is an inelastic collision

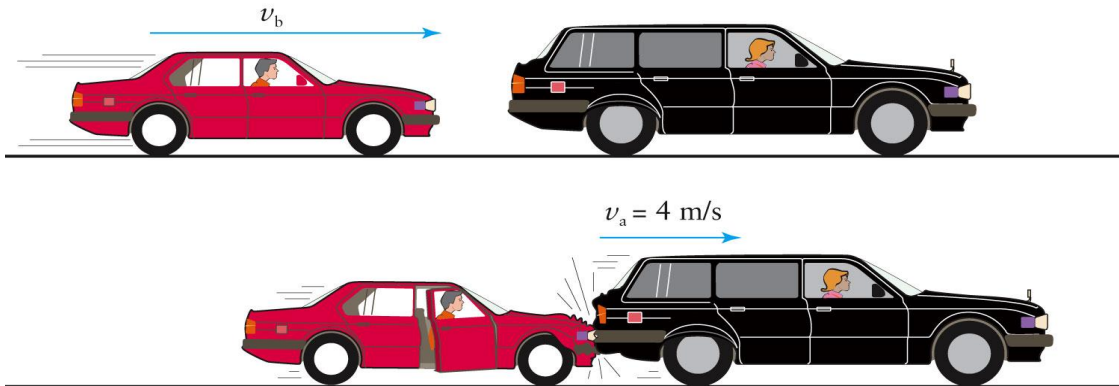


Example 3.10

Recall the automobile collision analyzed in Example 3.2. Compare the amounts of kinetic energy in the system before and after the collision:

$$KE_{\text{before}} = \frac{1}{2} \times 1,000 \text{ kg} \times (10 \text{ m/s})^2 = 50,000 \text{ J}$$

$$KE_{\text{after}} = \frac{1}{2} \times 2,500 \text{ kg} \times (4 \text{ m/s})^2 = 20,000 \text{ J}$$



Example 3.10

- 30 kJ of energy was lost during the collision, converted to heat and sound, used to deform the cars, etc.
- Since KE was lost, the collision was inelastic. It is perfectly inelastic because they stuck together.

3.7 Power

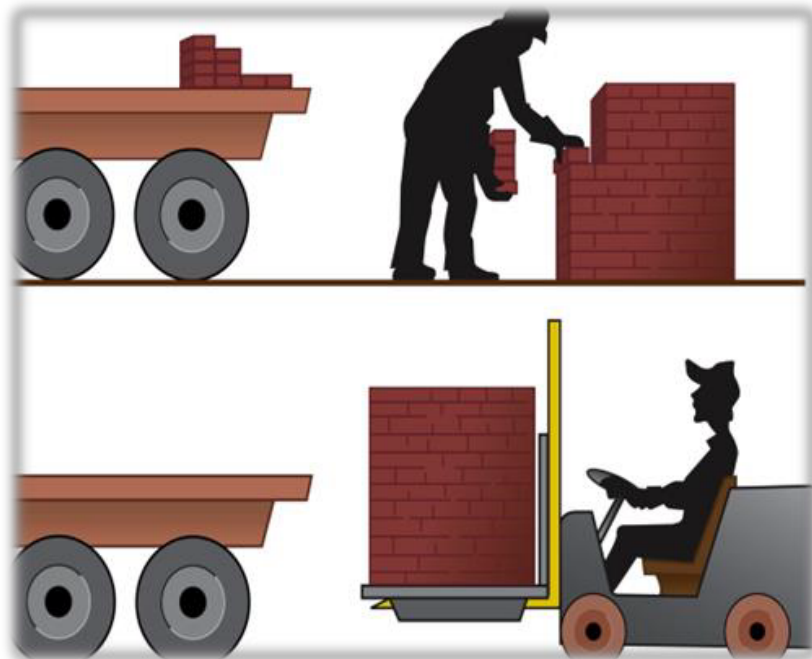
If a ton of bricks needs to be loaded from the ground onto a truck. This might be done in one of two ways.

First, a person could lift the bricks one at a time and place them on the truck.

This might take the person an hour.

Second, a forklift could be used to load the bricks all at once.

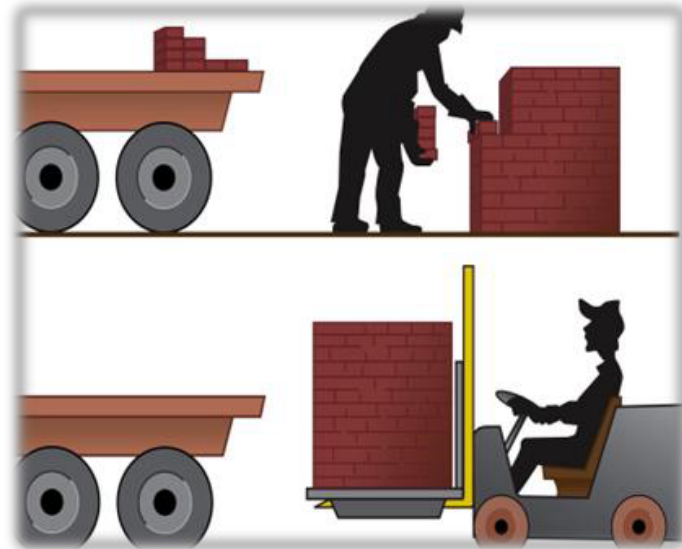
This might take only 10 seconds.



3.7 Power

In both cases, the same amount of work is done. The force on each brick (its weight) times the vertical distance it is moved (the height of the truck bed) is the same whether the bricks are loaded one at a time or all at once.

The work done, Fd , is the same in both cases, but the **power** is different.



3.7 Power

DEFINITION

Power: The rate of doing work. The rate at which energy is transferred or transformed. Work done divided by the time. Energy transferred divided by the time.

$$P = \frac{\text{work}}{t} \quad P = \frac{E}{t}$$

Physical Quantity	Metric Units	English Units
Power (P)	watt (W)	foot-pound/second (ft-lb/s)
		horsepower (hp)

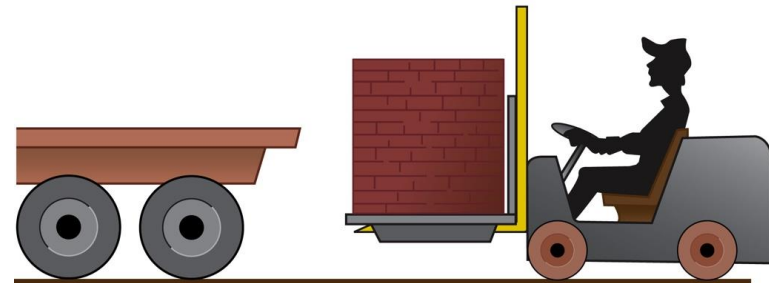
3.7 Power

In SI units, the weight of a ton of bricks is 8,900 newtons. If the height of the truck bed is 1.2 meters, then the work done is:

$$\begin{aligned}\text{work} &= Fd \\ &= 8,900 \text{ N} \times 1.2 \text{ m} \\ &= 10,680 \text{ J}\end{aligned}$$

$$\begin{aligned}P &= \frac{\text{work}}{t} = \frac{10,680 \text{ J}}{10 \text{ s}} \\ &= 1,068 \text{ J/s} = 1,068 \text{ W}\end{aligned}$$

$$1 \text{ watt} = \frac{1 \text{ joule}}{1 \text{ second}} \rightarrow 1 \text{ W} = 1 \text{ J/s}$$



Example

In Examples 2.2 and 3.5, we computed the acceleration, force, and work for a 1,000-kilogram car that goes from 0 to 27 m/s in 10 seconds. Let us now determine the required power output of the engine.

$$1 \text{ hp} = 550 \text{ ft-lb/s} = 746 \text{ W}$$

$$P = \frac{\text{work}}{t} = \frac{364,500 \text{ J}}{10 \text{ s}} = 36,450 \text{ W} = 48.9 \text{ hp}$$

Example 3.6

On 12 June 1979, Bryan Allen flew the *Gossamer Albatross* across the English Channel turning the propeller of the aircraft using a bicycle-like mechanism. The flight lasted 2.82 hours. If Allen produced a steady 250 watts of power during the flight, how much work did he do to complete the trip?

$$\begin{aligned}\text{work} &= Pt = 250 \text{ W} \times 2.82 \text{ h} \\ &= 250 \text{ W} \times 10,140 \text{ s} \\ &= 2,535,000 \text{ J} = 606 \text{ Calories}\end{aligned}$$



3.7 Power

There is usually no limit to how much work a device or a person can do.

But there is a limit on how fast the work can be done.

The power is limited.

3.8 Rotation and Angular Momentum

The final conservation law applies to rotational motion

LAWS

Law of Conservation of Angular Momentum: The total angular momentum of an isolated system is constant.

For a system to be isolated so that the law of conservation of angular momentum applies, the only net external force that can act on the object must be directed toward or away from the center of the object's motion.

The centripetal force required to keep an object moving in a circle fits this condition.

Rotation and Angular Momentum

Angular momentum is the momentum of an object moving in a circle.

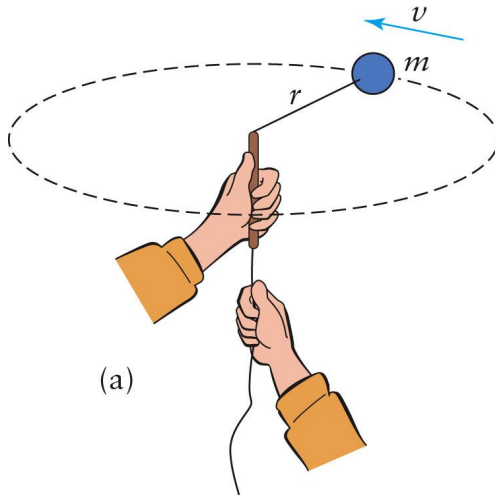
$$\text{angular momentum} = mvr \quad (\text{circular path})$$

The angular momentum of the object is also equal to its linear momentum (mv) multiplied by the radius of its circular path.

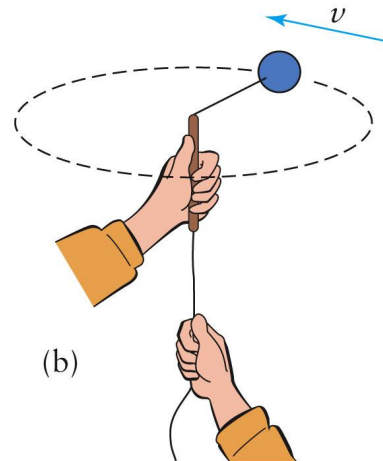
like linear momentum, angular momentum is a vector.

3.8 Rotation and Angular Momentum

To illustrate the law of conservation of angular momentum, imagine a ball circling overhead on the end of a string that passes through a tube. *no friction or air resistance*



(a)



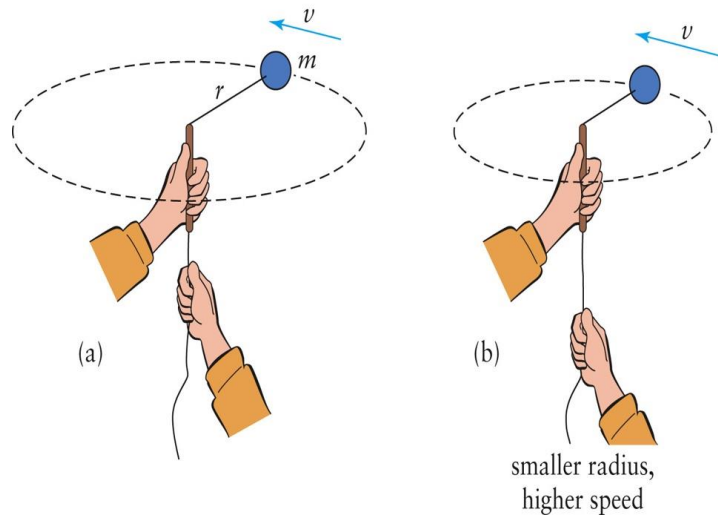
(b)

smaller radius,
higher speed

The faster the ball goes, the greater its orbital angular momentum.

3.8 Rotation and Angular Momentum

If we shorten the string, the object's speed increases. If you let the string out so the radius is larger, the ball will slow down, keeping the angular momentum constant.



As force exerted on the ball is directed toward the center of its motion, angular momentum is conserved—it has the same value before and after the change.

3.8 Rotation and Angular Momentum

You spin a ball attached to the end of a 1.0 meter string with a speed of 10 m/s. Find the ball's speed as you shorten the string to 10 centimeters.

$$v_i r_i = v_f r_f \quad \Rightarrow \quad v_f = \frac{r_i}{r_f} v_i$$

$$\begin{aligned} v_f &= \frac{r_i}{r_f} v_i = \frac{1.0 \text{ m}}{0.10 \text{ m}} \left(10 \frac{\text{m}}{\text{s}} \right) \\ &= 100 \text{ m/s} \end{aligned}$$

Important Equations

IMPORTANT EQUATIONS

Equation	Comments	Equation	Comments
<i>Fundamental Equations</i>		<i>Special-Case Equations</i>	
linear momentum = mv	Definition of linear momentum	$v = \sqrt{2gd}$	Speed after falling freely a distance d
$F = \frac{\Delta(mv)}{\Delta t}$	Alternate form of Newton's second law	$d = \frac{v^2}{2g}$	Height reached given initial speed v
$\Delta(mv) = F \Delta t$	Impulse-momentum relation	angular momentum = mvr	Angular momentum (circular motion)
work = Fd	Definition of work	torque = $\frac{\Delta(mvr)}{\Delta t}$	Newton's second law for rotation (circular path)
$KE = \frac{1}{2} mv^2$	Kinetic energy		
$PE = Wd = mgd$	Gravitational potential energy		
$PE = \frac{1}{2} kd^2$	Elastic potential energy		
$P = \frac{\text{work}}{t}; \quad P = \frac{E}{t}$	Definition of power		